Predicting the Zero-Shot Transfer of a Promptable Video Tracker

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Abstract

Conservation teams increasingly monitor wildlife with camera traps, which record far more footage than anyone can watch. Locating each animal in a video and following it from frame to frame is the bottleneck, and a new generation of promptable trackers promises to remove it: name an animal in ordinary words and the model finds and follows every one that appears, without ever having been trained on that species. SAM 3 is the current example. The difficulty is that this ability is uneven in ways nobody can anticipate — strong on one animal or location, poor on the next — so a team pointing such a model at a new species has no way of judging beforehand whether to trust its output or fall back on manual annotation.

This dissertation asks whether that judgement can be made in advance. It tests a natural intuition: that a model should struggle most on the animals and places least like those it has seen before. Were that true, one could measure how unfamiliar a species or a setting is — cheaply, without labelling anything and without running the model at all — and use the measurement to forecast reliability. Four such measures of unfamiliarity are tested, and the question is put separately to two distinct abilities: finding the animal at all, and keeping the identities straight once it has been found.

The answer is that it does not work. On a large camera-trap benchmark of wild animals, no measure of unfamiliarity predicts performance on species or places the model has not seen. The only property that predicts anything is how large the animal appears in frame — a nuisance rather than a discovery — and a second attempt based on how difficult the footage looks, such as darkness and visual clutter, collapses into that same size effect. The finding is not an accident of one dataset or one model: it survives on a second, independent dataset and on two further trackers of quite different design. Nor is it a failure of method, since the same test reliably detects the size effect whenever it is present.

The contribution is therefore a carefully bounded negative result, the first for this class of model, and with it a practical caution: resemblance to familiar training data is no guarantee that such a tracker will work, and reliability has to be checked rather than assumed.

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I thank my supervisor, Dr Marwa Mahmoud, for guidance throughout this project.

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`pHOTA`	Promptable Higher Order Tracking Accuracy — the benchmark's tracking score
`pDetA`	Its *detection* half: were the right animals found? (our primary target)
`pAssA`	Its *association* half: was each identity kept over time?
HOTA	Higher Order Tracking Accuracy, the metric `pHOTA` derives from
IoU	Intersection over Union, the mask/box overlap measure
GLM	Generalised Linear Model (here a support-weighted logit-link regression)
CI	Confidence Interval (throughout, a 95% group-cluster-bootstrap interval)
MAE	Mean Absolute Error
ΔMAE	Out-of-sample MAE gain over a mean predictor; positive = better than guessing the mean
LSO / LLO	Leave-Species-Out / Leave-Location-Out cross-validation
VIF	Variance Inflation Factor, a collinearity diagnostic
LMG	Lindeman–Merenda–Gold, the dominance-analysis variance split
MMD	Maximum Mean Discrepancy, a distributional distance
OOD	Out-Of-Distribution
ATC	Average Thresholded Confidence, an after-running accuracy estimator
GT	Ground Truth
IR	Infrared (night-time camera-trap imagery)
RLE	Run-Length Encoding, the COCO mask format
LVIS	Large Vocabulary Instance Segmentation, the class vocabulary BURST uses

Chapter 1

Introduction

Wildlife video is recorded far faster than anyone can annotate it, and a new class of promptable tracker promises to close that gap by finding and following animals it was never trained on. Whether such a model can be trusted on a particular animal in a particular place is not something anyone can currently say in advance. This chapter sets out that problem, turns it into a research question and a set of testable hypotheses, states what the work contributes, and maps the rest of the dissertation.

1.1 Motivation

Finding an animal in a video and following it over time is a need shared by several fields. Ecologists use it to survey wild populations from camera-trap footage. Neuroscientists and behavioural biologists use it to quantify what an animal does frame by frame in the laboratory, where the tracked trajectory is often the measurement the science rests on, and where a whole tool-chain now exists to recover pose and turn it into behavioural labels [18, 21, 23]. The bottleneck is the same in both cases: video is cheap to record and expensive to annotate.

This dissertation works on the wild side of that divide, where the problem is hardest. Camera traps have become a standard tool in wildlife monitoring, and a single survey leaves a team with far more footage than anyone can watch. Somebody still has to work out which animal is in each clip, how many there are, and what they do. Done by hand this is slow and expensive, and it is the step that limits how much ecology gets done at all.

Automating it means solving two problems, not one. The first is *finding* the animal in a frame. The second is *following* it from frame to frame, so the same individual keeps the same identity for as long as it is visible. Only the second yields the counts and behavioural sequences both communities want: in a study of social behaviour, losing an identity means losing the very animal whose behaviour was being measured. It is also the harder of the two.

Neither is easy on real footage. Animals are camouflaged against the background they live in, appear at any distance and viewpoint, and are cut in half by vegetation. Much of the footage is recorded at night in infrared. And because a camera trap is fixed, the same scene repeats in every clip from that site, which gives a model an easy shortcut: learn the background rather than the animal [5, 33]. Section 2.2 sets these difficulties out in full.

For years the standard answer was to train a detector on a fixed list of species and deploy it. It works where it was trained. It degrades sharply at camera sites and regions it has not seen [3, 16]. And adding a species that was not on the list means collecting and labelling more data — precisely the cost a team opening a new site is trying to avoid.

Promptable models change that arrangement, because the prompt can now be a phrase. Give SAM 3 [7] `"impala"` and it finds, segments and follows every impala in the clip, having never been trained on that species. A team can point a tracker at an animal it has never seen and get an answer straight away.

That is the appeal. The liability is that the answer is not equally good everywhere. Zero-shot performance varies sharply from one animal to the next, and from one place to the next. No rule tells a practitioner, before running the model, whether its output can be trusted for a given animal in a given place. The choice left is an unattractive one: trust the output blindly, or fall back on the manual annotation the model was meant to replace.

This dissertation studies that gap. It asks whether such a rule exists — first for SAM 3, and then whether whatever we find is specific to it.

1.2 Problem statement and research question

The appeal of a zero-shot model is that it skips the labelling step. That is also what makes it hard to trust. The normal way to check whether a model works on your data is to label some of it and measure. But a team that could afford to do that would have had less need for a zero-shot model in the first place. So these tools get deployed precisely where nobody has checked that they work.

The problem is a general one. Any foundation model applied to a case nobody validated sits in the same position. The capability is broad, the guarantees are absent, and the user has no principled basis for deciding how far to trust the output. What would settle it is a rule that can be checked *in advance*: some property of the target, cheap to measure, that says whether the model will cope.

There is an obvious candidate, and practitioners already reason with it informally — distance from the familiar. A model ought to do worse on things unlike what it has seen. If that intuition held quantitatively, it would be genuinely useful. You could measure how far a new species or place sits from the model's reference. That number would then act as a deployment gate, applied before spending compute and before annotating anything.

This dissertation asks whether that rule exists. Precisely: can a promptable tracker's zero-shot accuracy on an unseen (species, place) be predicted in advance, from properties measurable without running the model or scoring its output? And which factors, if any, govern that transfer?

We ask it separately for the two halves of the task, because tracking is built on detection. The model must first find the animal, then hold onto its identity from frame to frame. The two can fail for different reasons. Finding is about whether the animal is recognisable at all; following is about whether the scene lets an identity survive. A single accuracy number would average them together and hide the answer, so we keep them apart throughout.

Asking the question at all requires a demanding dataset. It needs species labels structured enough to say how far apart two species are, plus location metadata, timestamps, and dense per-frame tracking annotation — all in one corpus. SA-FARI [30] carries all four, which is why we use it; Section 2.7.1 sets out its contents.

We study SAM 3 on SA-FARI as the primary case. We then ask whether the answer is specific to it. The analysis is repeated on a second, independent dataset, and two other promptable trackers are swapped in on the same data (Section 4.11).

1.3 Hypotheses

The research question asks two things at once: *whether* transfer can be predicted, and *what governs* it. The hypotheses below turn that into something a dataset can answer. Each names a factor, commits to a direction, and can be rejected by the evidence.

Two terms are needed first, because the rest of the dissertation rests on them.

The tracker is scored with a standard tracking metric that splits cleanly into two halves (Section 2.3). **pDetA**, detection accuracy, scores the finding half: were the right animals located? **pAssA**, association accuracy, scores the following half: did each animal keep one consistent identity for as long as it was visible? Both run from 0 to 1, and we model them separately for the reason given in Section 1.2.

A predictor is **label-free** if computing it needs no annotation of the target footage and no measurement of how the tracker performed on it. Only such a predictor can be evaluated *before* the model is run, which is exactly what a deployment gate would require. The four tested here are all *distances*: how far a target species or place sits from a fixed reference set, measured by taxonomy, by appearance, by the surrounding scene, and by time.

H0 (null)
: The label-free distances carry no predictive information about `pDetA` or `pAssA`. Transfer is not distance-driven.

H1 (detection)
: `pDetA` falls as the species becomes more novel — further from the reference in taxonomy and in appearance.

H2 (association)
: `pAssA` falls as the scene becomes harder — an unfamiliar background, night or infrared capture, visual clutter.

H1 and H2 divide the research question between the two halves of the task, and each attaches the half to the factor most likely to govern it. H1 is the one that matters most in deployment. If it held, a team could look up an unfamiliar species and know in advance whether detection is likely to hold up. H2 rests on a different intuition. Holding an identity across frames should depend on the scene rather than on which animal it is. If so, the difficulties catalogued in Section 2.2 — darkness, clutter, an unfamiliar background — ought to bite here rather than on detection.

H0 is what remains if both fail. It is not an absence of an answer but a substantive one. It would mean the distance intuition, however reasonable it sounds, does not survive contact with the data — and that a deployment gate cannot be built this way.

1.4 Contributions

To our knowledge this is the first study to ask whether a promptable video tracker's zero-shot transfer can be predicted before the model is run, from properties that require no labelling of the target, and the first to put that question to an out-of-sample test. Section 2.8 places that claim against the existing literature. The work contributes four things, taken up again in Section 6.2.

(i) A testable formulation of the question, and a validation protocol that any future before-running estimator can be held to.

(ii) The detection–association split brought to label-free evaluation, so that an answer can be attributed to finding an animal or to following it rather than averaged across both.

(iii) The answer itself, established at scope: label-free distance does not predict transfer, and the finding survives an independent second dataset and two architecturally different trackers.

(iv) An account of why the obvious candidate fails — the one apparent effect reduces to how large the animal appears in frame — and the caution for practice that follows.

1.5 Dissertation outline

Chapter 2 reviews the related work and states precisely the gap this dissertation fills. Chapter 3 describes the data and the two splits, how transfer is measured, the four label-free distances, and the statistical procedure used to test them. Chapter 4 reports the results, including the out-of-sample validation, the robustness checks, and the replications on a further dataset and two further trackers. Chapter 5 interprets those results, reconciles them with the prior literature, and sets out the limitations and threats to validity together with the ethical considerations. Chapter 6 concludes and identifies what is left open for future work.

Chapter 2

Background and Related Work

This chapter reviews the work this dissertation builds on. It covers the class of model under study, the difficulties that make animal detection and tracking hard in the first place, how tracking is scored, the literature on estimating a model's performance without labels, what is known about distribution shift in camera-trap vision, the representations that make a distance between species meaningful, and the datasets involved. It closes by stating the gap that remains.

2.1 Promptable and open-vocabulary segmentation and tracking

For most of the last decade, detection and segmentation models were trained against a fixed list of categories. Such a model cannot be asked about anything outside that list. Extending it to a new category means collecting labelled examples and training again. In camera-trap work the standard tool of this kind is MegaDetector [4], a detector trained on millions of camera-trap images to localise animals, people and vehicles, and widely deployed as a first-pass filter over survey footage.

The *Segment Anything* line broke the fixed-list assumption. It replaced the label set with a prompt: a user indicates what should be segmented, and the model returns a mask for it. SAM 2 [25] carried the formulation from images to video, propagating a mask across frames from a point or box placed in one of them.

SAM 3 [7] changes what the prompt may be. Instead of a point or a box it accepts a short noun phrase, and returns every instance matching that phrase, segmented and tracked across the clip. The category need never have been seen in training. This is a zero-shot, open-vocabulary detect–segment–track capability, and SAM 3 is the current state of the art for it.

Other models reach open-vocabulary video understanding by different routes. GLEE [31] trains a single unified model to detect, segment and track objects at scale, driven by a text query. Florence-2 [32] takes a caption-style approach, using one sequence-to-sequence model across a range of vision tasks and emitting regions in response to a task prompt, though without a confidence score attached to them. These are the main alternative designs in the same space.

2.2 What makes animals hard to detect and track

Wild footage is an unusually hostile input, and the difficulties are worth setting out because they are what any tracker deployed on it must survive.

The first is that animals are often hard to see at all. Many are camouflaged against the habitat they live in, and the resulting low contrast between object and background defeats methods that assume a visually salient foreground. This is a recognised problem in its own right: camouflaged object detection is studied as a separate task with dedicated benchmarks, precisely because standard detectors underperform on it [11].

The second is variability of appearance. A camera trap is triggered by whatever passes, so an animal may fill the frame in one clip and occupy a few dozen pixels in the next. It may face the lens or present only a flank, and vegetation frequently occludes part of it. A bounding box is a poor description of an animal in such conditions, which is one reason segmentation — labelling the pixels belonging to the animal — is the more informative output.

The third is image quality. Much wildlife footage is captured at night under infrared illumination, where colour information is absent and contrast is poor. Infrared and thermal imaging are established wildlife monitoring modalities with their own detection characteristics, and their properties differ enough from daylight imagery that observation strategies are designed around them [6]. Motion blur is common, since animals are typically moving when they trigger the camera.

The fourth is context. A camera trap is fixed, so the same scene recurs in every clip from that site, and a model can come to rely on the scene rather than on the animal. This failure mode has a literature of its own, reviewed in Section 2.5.

The fifth applies to tracking rather than detection. Keeping an identity attached to the right animal is a distinct problem from finding it. Animals leave the frame and re-enter it, pass behind vegetation, and in groups may be near-indistinguishable from one another. Identity errors are treated as a first-class failure mode in multi-object tracking benchmarks, which annotate visibility explicitly and report identity switches separately from misses [9]. Recovering the identity of a specific individual across sightings is hard enough in wildlife imagery to constitute its own research area [27].

2.3 Evaluating multi-object tracking

Multi-object tracking is scored by comparing predicted tracks against annotated ones. The difficulty is that two very different failures — missing the object, and swapping its identity — both reduce a single accuracy figure, and earlier metrics combined them in ways that were hard to interpret.

Higher Order Tracking Accuracy (HOTA) [17] was introduced to separate them. At each localisation threshold α it forms a detection term (`DetA`: were the right objects found?) and an association term (`AssA`: were identities kept consistent over time?), then combines them as a geometric mean:

$$\mathrm{HOTA}_\alpha = \sqrt{\mathrm{DetA}_\alpha \cdot \mathrm{AssA}_\alpha}, \qquad \mathrm{pHOTA} = \frac{1}{|\mathcal{A}|} \sum_{\alpha \in \mathcal{A}} \mathrm{HOTA}_\alpha, \tag{2.1}$$

averaged over the threshold set $\mathcal{A} = \{0.05, 0.10, \ldots, 0.95\}$. Because the two terms are formed separately before being combined, a HOTA score can be read either as a single number or as two.

SA-FARI scores promptable tracking with `pHOTA`, a promptable variant that decomposes identically into `pDetA` and `pAssA`.

2.4 Predicting model performance without labels

A now-mature literature shows that a model's performance under distribution shift can be anticipated without labelling the target data.

Miller et al. [20] observe that out-of-distribution (OOD) accuracy is strongly linearly correlated with in-distribution accuracy, across many models and many shifts. Baek et al. [2] show that the agreement between a pair of models tracks OOD accuracy closely enough to estimate it from unlabelled data alone. A parallel strand reads the model's own statistics on the unlabelled target. Average thresholded confidence [13] calibrates a confidence threshold on source data and reads off target accuracy as the mass above it, while Deng and Zheng [10] regress classifier accuracy directly on a distribution-shift feature — the Fréchet distance between train and test feature sets.

The idea has only recently reached structured outputs. Yoo et al. [34] estimate object-detection performance without ground truth, from the spatial consistency and confidence reliability of pre-NMS boxes. Reverse classification accuracy [29] predicts per-image segmentation quality by training a reverse model against a labelled reference.

Two properties characterise essentially all of this work. First, the target is an image classifier scored by a single scalar, or at most a detector or segmenter operating on a static image. None of it concerns identity maintained over time, and none decomposes the estimate into separate components. Second, every one of these methods runs the model on the target and inspects what comes out — its predictions, its confidences, or its agreement with another model. Each estimates the score of a model that has already been executed.

2.5 Distribution shift in camera-trap and wildlife vision

That camera-trap models degrade at new locations is well established. *Terra Incognita* [3] showed that recognisers excellent at trained camera sites fail sharply at unseen ones. WILDS [16] consolidated this, via iWildCam, as a canonical real-world shift. PanAf-FGBG [5] demonstrated that behaviour-correlated backgrounds causally drive the drop, and proposed a latent-space mitigation.

The claim that scene context, rather than the object, drives such failures has a longer pedigree in general vision. Xiao et al. [33] isolate foreground from background on ImageNet, in the ImageNet-9 "Background Challenge", and show that a classifier scores well above chance from the background *alone*. An adversarially chosen background flips a correctly classified foreground up to 87.5% of the time. Background is genuine signal, not noise. Rosenfeld et al. [26] make the parallel point for detectors: transplanting an object into an unfamiliar scene suppresses its detection, flips its predicted identity as it moves, and disturbs other detections in the image, exposing a reliance on context and co-occurrence.

Xiao et al. [33] also report that more accurate and more robust models rely on background *less*, which suggests the effect is not fixed but a property of how strong the model is.

What this literature shares is its direction of travel. It measures degradation after the fact, or intervenes on the input to mitigate it. It does not forecast per-site or per-species reliability in advance. It also studies classifiers and behaviour recognisers rather than promptable trackers, scored by a single accuracy with no detection/association split.

2.6 Representations of species and scenes

Measuring how far one species sits from another, or one scene from another, requires a representation in which such a distance means something.

Vision foundation models turn out to carry that structure. BioCLIP [28] shows that a model trained on a large biological image corpus encodes taxonomic structure across the tree of life, its feature space reflecting the hierarchy that runs from kingdom down to species. Appearance and phylogeny are correlated, so a representation that separates species visually will tend also to order them taxonomically.

General-purpose encoders offer the same service without biological supervision. DINOv2 [22] learns visual features self-supervised, and they transfer across domains without fine-tuning. CLIP [24] aligns images with natural-language descriptions, producing a space in which images and text share coordinates. Both are widely used as off-the-shelf feature extractors.

2.7 Datasets

Two datasets are central to this work: SA-FARI, on which the primary study runs, and BURST, which carries the independent replication.

2.7.1 SA-FARI

SA-FARI [30] is a large open multi-animal camera-trap tracking dataset, released alongside SAM 3. It contains 11,609 densely annotated videos spanning 99 species, recorded at 741 locations across four continents. Every species carries a seven-level biological taxonomy running from kingdom to species, and every video carries a creation timestamp. Annotations are per-frame masks with persistent identities, and the benchmark ships an official evaluator, `VEval`.

Two features set it apart. The taxonomy makes the relationship between two species measurable rather than a matter of judgement, and the location and timestamp metadata do the same for place and era. The benchmark also includes abundant *hard negatives*: prompts naming a species that is absent from the clip, which a correct run should answer with nothing. These make it possible to study false alarms alongside misses.

SA-FARI ships a train/test partition drawn along locations, so the test camera sites are unseen while most species occur on both sides of it.

2.7.2 BURST

BURST [1] is a video benchmark for object recognition, segmentation and tracking, built on the TAO [8] video corpus. It annotates 482 categories from the LVIS [14] vocabulary with per-frame masks and persistent track identities, and a portion of that vocabulary is animal categories. Its footage comes from internet, driving and film sources rather than camera traps, and it carries neither location nor timestamp metadata. It is mask-native and considerably more varied in subject matter than SA-FARI, which makes it a useful independent point of comparison.

2.8 Summary and gap

Predictable OOD generalisation is established for classifiers scored by a scalar [2, 10, 13, 20] and, very recently, for static detectors and segmenters [29, 34]. Camera-trap vision has thoroughly documented, but not forecast, location-driven degradation [3, 5, 16]. Most tellingly, a recent unified study of camera-trap recognition over time [15] explicitly names "predicting when zero-shot models will succeed at a new site" as an open question, and stops at naming it.

No prior work predicts, *before running the model* and from label-free distances, the zero-shot transfer of a promptable video tracker. None separates that prediction into detection and association. That is the gap this dissertation addresses.

Chapter 3

Data and Methodology

This chapter sets out how the question posed in Chapter 1 was turned into something a dataset can answer. It begins with the research design, the terms that design depends on, and the rule by which a predictor counts as validated. It then describes the data and the two splits, how the tracker's performance is measured, and how the label-free features are computed against a frozen reference. The remaining sections cover the statistical model and the tests applied to it, the replication on a second dataset and a second tracker, and the constraints that make the work reproducible.

3.1 Research design and evaluation criteria

This is a predictive-modelling study. On one side is a set of predictors X: four distances, each measuring how far a target species or place sits from a frozen reference, none of them requiring the tracker to be run or the target to be labelled. On the other is a target Y: the tracker's per-cell detection and association scores. The question is whether X predicts Y, and which parts of it do the work, asked separately for the two halves of the task. Nothing here attempts to make the tracker score higher — it is the fixed object of study. Three things need pinning down first: what running it zero-shot means, what the two scores are, and what would count as having predicted them.

3.1.1 What "zero-shot" means here

A model is run zero-shot on a category when it is asked to find that category without having been trained on labelled examples of it, and without any change to its weights at the time of asking. The only thing telling it what to look for is the prompt. A conventional detector, by contrast, is trained on a fixed list and can be asked about nothing else; a fine-tuned model has had its weights updated on target-domain data; a few-shot model is handed a small number of labelled examples. A zero-shot model gets the word `"impala"` and nothing more.

Three commitments follow and hold throughout. The tracker is the public released checkpoint, and its weights are never updated — not fine-tuned, not adapted, not calibrated on any part of this data. The only input identifying the target is the text prompt. And no labelled example of the target species ever reaches the tracker: the ground-truth masks are read afterwards, to score what it produced and to crop images when building the distance features (Section 3.4), never to tell it what to look for. The only thing fitted anywhere in this study is the small regression mapping distances onto scores. Note that zero-shot is meant relative to the dataset and to the reference frozen inside it, not as a claim about SAM 3's undisclosed pretraining corpus — a scope limit taken up in Section 5.4.

3.1.2 What is being predicted: finding and following

The evaluator compares what the tracker produced against what a human annotated. Both sides consist of masklets, a masklet being one object's mask followed through the clip frame by frame, and a predicted mask counts as matching an annotated one when their intersection over union reaches a threshold α. Given that matching, `pDetA` (detection accuracy) asks whether the right animals were found, rewarding annotated animals that were located and penalising both animals missed and masks invented where no animal was. `pAssA` (association accuracy) asks whether identities were kept, measuring across the matched detections how consistently one predicted identity corresponds to one annotated identity over time; it falls when one animal's track is split among several predicted identities, when two identities are swapped, and when two animals are merged into a single track.

Both lie in $[0, 1]$, higher is better, and both are averaged over $\alpha \in \{0.05, \ldots, 0.95\}$, so neither depends on one arbitrary choice of how much overlap is close enough (Equation 2.1). The mask-based variants are used throughout, SA-FARI being mask-annotated, and neither metric is reimplemented: the official `VEval` evaluator is run as supplied and its output read.

The two are not symmetric, and the asymmetry matters. Association is conditional on detection, since an identity can only be assessed for an animal that was found in the first place. A second consequence is worth flagging at the outset. Where a clip holds a single animal there is only one identity to keep, so `pAssA` is then largely determined by `pDetA` and carries little variance of its own. How often that happens is a property of the data rather than of the design, and Chapter 5 reports it; where it dominates, association is reported as the weak target it is rather than as an independently powered test. The external replication of Section 3.6 was chosen partly because it does not have this problem.

3.1.3 Definition of success and the decision rule

The intended deliverable is a validated, out-of-sample predictor of `pDetA`/`pAssA` from label-free distances, with confidence intervals that respect the group structure (Section 3.5.2). A distance is taken to predict transfer only if it clears both bars:

(i) its coefficient's group-cluster-bootstrap 95% confidence interval (CI) excludes zero; and

(ii) the grouped cross-validated model beats a mean-predictor baseline by a margin that is significant under a paired group-bootstrap.

The bars are not redundant. Criterion (i) says a relationship is visible in the data the model was fitted on; it does not say that relationship holds for a species or place never seen, which is the entire practical claim. Criterion (ii) is that test, and its baseline is deliberately humble — predicting each held-out cell with the training mean, which is what a practitioner with no predictor would effectively assume. A feature clearing (i) but not (ii) has been fitted, not validated.

H1 (detection $\leftarrow$ species novelty) and H2 (association $\leftarrow$ environment difficulty) are supported only when their respective distances meet this bar on the split that isolates them. If neither does under tests built deliberately to be powerful, the conclusion is H0 — distance explains little variance — which is a substantive result rather than a failure to find one, and motivates the representational-probing direction of Chapter 5.

3.2 Data, splits, and the analysis unit

SA-FARI [30] is the primary dataset, and everything in this section describes it. The independent replication runs on a second dataset, BURST, which Section 3.6 covers separately.

Each test is a probe: one video paired with one text prompt. The prompt names a species, which resolves its taxonomy, and the video supplies the location and the timestamp. Probes are then aggregated into cells, a cell being one (species, location, time) group. Each cell carries support counts (n_{frames}, n_{masklets}, n_{videos}), used for weighting and as covariates. The cell is the unit of analysis throughout: it is what carries a score, and what the regression is fitted on.

Hard negatives — prompts naming a species that is absent — are kept, because they carry the false-positive side of detection. There is one exception. The large pooled inference pass uses a stratified per-species cap to remain feasible, and under that cap hard negatives are set aside from inference and analysed separately instead (Section 3.5.6).

Two complementary splits. SA-FARI's shipped split separates locations but not species (Section 2.7.1). Species-distance is therefore ≈ 0 for nearly every cell, so that split cannot test species novelty on its own. Because the tracker is frozen (Section 3.1.1), the shipped split plays no training role and serves only to define our reference distribution, which leaves us free to repartition the same videos. Inference is run once — scoring is per-video and split-agnostic — and two splits are overlaid on the result:

Split A — species hold-out (primary).
: Species from both official splits are pooled and held out one at a time, so a held-out species' distance is measured to the nearest remaining species. Species-distance varies while the environment stays familiar, which isolates H1.

Split B — location hold-out (secondary).
: The official split, whose unseen locations isolate H2, and whose comparability with the published benchmark carries the initial score-sanity check.

Split A is a constructed hold-out rather than a shipped one. That is legitimate precisely because the model is frozen, and it is reported as constructed throughout.

Split B yields 1,447 scored cells, spanning 113 taxonomy-bearing categories and 92 unseen test locations. Of those cells 346 are positive, meaning the queried species is present so that `pDetA` and `pAssA` are defined; the remaining 1,101 are hard negatives, drawn from 1,486 hard-negative probes. Split A pools the species present in both official splits into 1,025 scored cells.

3.3 Measuring transfer

Figure 3.1 sets out the pipeline as a whole. This section describes its upper path: how the frozen tracker is run and scored into the target Y. Sections 3.4 and 3.5 describe the lower path and the modelling.

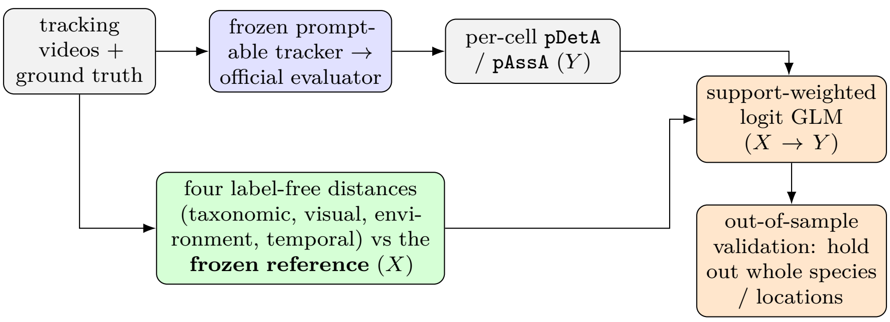

Figure 3.1: The measurement-and-modelling pipeline, identical for every dataset and tracker. *Top:* the frozen tracker is scored by the official evaluator into per-cell detection/association scores (the target Y). *Bottom:* four label-free distances are computed against a frozen reference (the predictors X). A support-weighted logit GLM relates X to Y, validated out of sample by holding out whole species and locations. Only the GLM is fitted; the tracker is never trained.

The primary case is SAM 3 on SA-FARI, and the rest of this section gives its concrete choices and numbers. The replication and the model swap (Sections 3.6–3.6.1) later re-run the same pipeline on other datasets and trackers.

SAM 3 [7] is run frozen over the probe videos and scored with the official `VEval` evaluator, which yields a per-cell `pDetA`, `pAssA` and `pHOTA` [17] together with its support. Species-specific prompts are the primary condition, and a generic "animal" prompt is run as a robustness check. One inference pass over the union of the Split A and Split B probes serves both experiments. Predictions are keyed per (video, species), so that the several prompts issued against a single video cannot collide, and the harness can resume video by video.

The frame-level scores are aggregated two ways, and the two are kept distinct. The micro score is the dataset-level `VEval` figure, reported only to confirm that we reproduce the published SA-FARI operating point of ≈ 0.65 mask-`pHOTA`. The macro score is the support-weighted mean over positive cells, and it is the modelling target: the model is fitted cell by cell, so the cell is what must carry a score. Being a positives-only per-cell average, the macro mean sits lower, at ≈ 0.53. Unless a figure is explicitly identified as the micro score, every measurement number quoted in this dissertation is the macro value.

Validity of the frozen protocol. Because the tracker is frozen throughout (Section 3.1.1), reusing that single inference pass across both splits is sound rather than a leak: a cell's score cannot depend on which reference partition is later overlaid to compute its distances.

Freezing also settles the benchmark's status. SA-FARI is a held-out, zero-shot benchmark for the released SAM 3. The SAM 3 report [7] does not list SA-FARI among its training sources, and the SA-FARI paper [30] reports the released checkpoint zero-shot, alongside separately fine-tuned rows whose gain we deliberately forgo. Non-membership can therefore be verified at the dataset level. It cannot be verified frame by frame, because SAM 3's full pretraining corpus is undisclosed. That is why every distance is measured against the SA-FARI reference rather than against whatever SAM 3 actually saw, and why the familiarity proxy of Section 3.4.6 is included as a hedge.

3.4 Label-free distance features

The features defined here fall into three groups. Four are distances, one for each axis along which a target can be unfamiliar: taxonomic, visual, environment and temporal. These four are the hypothesis variables, with H1 resting on the first two and H2 on the third. The second group holds a single nuisance covariate, animal size. The third holds the familiarity proxy.

For each split, the reference ("seen") species, locations and taxonomy are frozen once, and every distance is then computed against that reference alone. This is the leakage firewall: no information about a probe can enter a feature. A unit test verifies it, swapping the reference and checking that the distances move as expected. We assert the axis that is genuinely disjoint — location for Split B, the held-out species for Split A — and report any residual species overlap rather than asserting it away.

Because the distances are computed against that reference, they require no measurement of the tracker's performance on the target: no `pDetA`, no `pAssA`, no annotation of how well SAM 3 did. That is the sense in which they are label-free, and it is the sense the research question needs, since it makes them available before the model is run.

One qualification belongs here rather than in a footnote. As implemented on an annotated benchmark, the visual, environment and size features read the dataset's ground-truth masks to separate animal from scene. That is a convenience of working on SA-FARI, not a requirement of the method: in deployment the same species prototype would be built from the handful of exemplar images a practitioner already has for a named species. It does bound the claim, and Section 5.4 returns to it.

3.4.1 Taxonomic distance

This distance measures how far apart two species sit on the tree of life.

Every species carries a seven-level path from kingdom down to species. Write that path $\pi(s)$, and let shared(a, b) count the levels on which two paths agree, starting from the top. Two species are then as far apart as the number of levels they do not share. A cell's distance is the smallest such value over the reference set $\mathcal{R}$:

$$\tau(a, b) = L - \mathrm{shared}(a, b) \quad (L = 7), \qquad d_{\mathrm{tax}}(s) = \min_{s' \in \mathcal{R},\, s' \neq s} \tau\big(\pi(s), \pi(s')\big). \tag{3.1}$$

So a species sharing a genus with its nearest reference neighbour scores 1, a shared family 2, and a shared order 3. A species sharing no level at all would score 7. The exclusion $s' \neq s$ is the leave-species-out rule, which keeps the measure non-trivial for species present on both sides of a split. Species whose taxonomy is incomplete are left undefined rather than imputed.

3.4.2 Visual distance

This distance measures how different one species looks from another.

Each animal is cropped to its ground-truth mask, with the background zeroed, and embedded with DINOv2 [22]. A species is then summarised by a single prototype: the mean of its embeddings, renormalised to unit length. The distance from an embedding v to the nearest prototype other than its own is one minus the largest cosine similarity,

$$p_j = \frac{\bar{v}_j}{\|\bar{v}_j\|_2}, \qquad d_{\mathrm{vis}}(v) = 1 - \max_{j \neq e} v^\top p_j, \tag{3.2}$$

where e indexes the query's own species, excluded again under the leave-species-out rule. A CLIP [24] encoder replaces DINOv2 as a robustness check. The masks come from the annotations (Figure 3.2b).

3.4.3 Environment distance

This distance measures how different one scene looks from another.

The animal is erased from the frame, its pixels replaced by the mean colour of what remains, and the result is embedded exactly as above (Figure 3.2c). Here prototypes are formed per location rather than per species, and d_{env} is Equation 3.2 applied to the nearest other location.

Three interpretable covariates come from the same frames. Infrared footage is greyscale, so a pixel's spread across colour channels, $\max_c f_c - \min_c f_c$, collapses towards zero; a frame counts as achromatic when that spread averages below 12. The low-light fraction is the share of a location's frames that are achromatic, and the night/IR flag is that fraction exceeding one half. Clutter is the mean variance of a Laplacian over the greyscale background, which measures local edge and texture density rather than colour.

3.4.4 Temporal gap

This distance measures how far apart in time two recordings were made.

Years come from the per-video creation-datetime metadata. For a cell recorded in year y, the gap is the smallest absolute difference to any reference year in $\mathcal{Y}$:

$$d_{\mathrm{temp}} = \min_{r \in \mathcal{Y}} |y - r|. \tag{3.3}$$

The feature depends on timestamps being both present and varied across cells. How far the available metadata actually supports it is reported in Chapter 4.

(a) frame + ground-truth mask (b) visual: mask-cropped animal (c) environment: animal erased

Figure 3.2: The two image-based distances on a camera-trap frame (a collared peccary). **(a)** ground-truth mask (orange); **(b)** the mask-cropped animal the visual distance fingerprints; **(c)** the animal mean-inpainted, which the environment distance fingerprints. None of these features runs SAM 3.

3.4.5 Animal size (confound control)

This covariate measures how large the animal appears in frame.

Let a be a species' mean ground-truth mask area in pixels, averaged over sampled masklets. The covariate is $\ln(1+a)$. Larger animals are easier to segment, so this is what separates genuine visual novelty from mere conspicuousness. Section 3.5 describes how it is used to test whether an apparent novelty effect is really a size effect.

3.4.6 SAM 3 familiarity proxy

This feature measures how familiar a species looks to SAM 3 itself.

Every distance above is measured against the SA-FARI reference rather than against what SAM 3 actually saw (Section 3.3). The proxy hedges that gap by reading familiarity from the model. The same mask-cropped animals used for the visual distance are pushed through SAM 3's image encoder, and the patch-token features are mean-pooled into a 1024-dimensional vector. Each species is then scored three ways: by silhouette, comparing own-cluster compactness against the nearest other species; by nearest prototype, which is Equation 3.2 recomputed with SAM 3's encoder in place of DINOv2; and by Mahalanobis typicality against the reference feature Gaussian.

The proxy stays before-running and label-free. It is a property of the images, not a function of SAM 3's tracking output, so it is not circular. Each of the three metrics is validated at the same leave-species-out bar, Bonferroni-corrected across them, and size-controlled. The decisive question is whether any of them adds out-of-sample gain over the visual distance and size together (Section 4.8).

3.5 Statistical modelling and hypothesis testing

Section 3.3 produced the target Y, and Section 3.4 the predictors X. This section relates the two, and puts the decision rule of Section 3.1.3 into practice.

Its order follows that rule. The first subsection fits the model, the second derives the confidence intervals that criterion (i) tests, and the third the out-of-sample comparison that criterion (ii) tests. The fourth divides explained variance between predictors that overlap. The fifth and sixth put two further questions that the four distances do not reach: whether intrinsic difficulty predicts better than novelty, and whether novelty predicts false alarms rather than misses. The last checks that the outcome does not hinge on any single design choice.

3.5.1 Per-target regression

Detection and association are modelled separately, because their decomposition is the intellectual core of the project. Section 3.1.2 gives the qualification. Association may carry little variance independent of detection, in which case it is reported as the weaker target rather than as an independently powered test.

Each target is bounded in $[0, 1]$. Each is therefore modelled with a support-weighted logit-link, or fractional-logit, regression:

$$\operatorname{logit} \mathbb{E}[y] = x^{\top}\beta, \qquad \text{fit with variance weights } w = n_{\text{frames}}, \tag{3.4}$$

where y is the cell's score and x the standardised design. That design carries the four distances, the scene covariates, and a $\log(n_{\text{frames}})$ term. The last of these is what keeps "rare" from being mistaken for "far".

3.5.2 Group-aware confidence intervals

A naive model interval would be far too narrow here, for two reasons. Support-weighting by frame count inflates the effective sample size. And a distance is constant within a species or a location, so hundreds of cells represent only tens of independent groups, which is pseudo-replication. We therefore resample whole species and whole locations, refit on each resample, and take percentile intervals. Every coefficient interval reported in this dissertation is that group-cluster-bootstrap interval.

3.5.3 Out-of-sample validation and the significance test

Validation holds out whole groups: leave-species-out on Split A, leave-location-out on Split B. The model is fitted without a group, then predicts it. Criterion (ii) compares the resulting error against the mean-predictor baseline, and a paired group-bootstrap resampling whole held-out groups tests that comparison. It returns both a confidence interval and a one-sided p-value on the gain.

Distinguishing absence of signal from absence of power. A null is only informative if the test could have detected an effect had one been present, so the design carries a positive control. The animal-size covariate is put through the identical procedure — the same clustered interval, the same whole-group hold-out — and the estimation and cross-validation machinery is unit-tested to recover planted effects on synthetic data. If the procedure clears the bar for size while staying flat for the distances, a flat result is evidence of absence of signal rather than absence of power. Chapter 4 reports whether it does.

3.5.4 Variance attribution and confound control

Four univariate fits would not show how much each factor explains, because the distances overlap one another. Variance is therefore attributed by dominance analysis, which divides shared explanatory power between correlated predictors, with a VIF check for collinearity. Raw correlations are reported alongside as a model-free cross-check. Where a distance does reach significance, the model is refitted with the animal-size covariate, to test whether the effect is size rather than novelty.

3.5.5 The intrinsic-difficulty pivot and the nested decomposition

The novelty distances are not the only label-free properties available. A complementary question, taken up after the novelty tests were complete, is whether transfer is governed by the intrinsic difficulty of the target imagery rather than by its novelty. The scene properties already measured for the environment axis — the low-light fraction, the clutter score and the night/IR flag — are promoted from nuisance covariates to predictors of interest, and tested against the same out-of-sample bar.

Difficulty and size are easily confused, so they are separated by a nested sequence of models: support only, size only, difficulty without size, and difficulty plus size. Comparing their out-of-sample gains exposes any apparent difficulty effect that is really size (Table 4.4). Each scene property's cell-level correlation is also reported beside its location-level value, because aggregating inflates correlations. That is the ecological-correlation fallacy.

3.5.6 Detection precision: false positives on hard negatives

The `pDetA` model conditions on the animal being present, so it measures recall given presence rather than hallucination. Hard negatives supply the other side. There the queried species is absent, so a correct run returns no masklet at all.

The predictions are read directly, and any returned masklet scoring above a threshold is flagged as a false positive. Each species' taxonomic and visual distance is attached, and the test asks whether the hallucination rate rises with novelty. It is run at the species level rather than the cell level, to avoid pseudo-replication.

3.5.7 Design-robustness checks

To test whether the outcome depends on any single design choice, the whole hold-out procedure is re-run with one component of the pipeline swapped at a time. The visual encoder changes from DINOv2 to CLIP. The crop changes from the mask-cropped animal to a whole-frame box that keeps the background. The prompt changes from species-specific to a generic "animal". And the visual-distance functional changes from a nearest-prototype cosine to a distributional Fréchet or MMD distance between the target and reference embedding sets, after AutoEval [10]. Each variant is fitted and validated at the identical whole-group bar, with significance judged after a Bonferroni correction over the variant family (Table B.5).

3.6 Independent replication on an external dataset

Every SA-FARI result reuses one frozen inference pass on one dataset. The two splits are therefore near-orthogonal, but not statistically independent. To test whether the outcome is a property of the transfer problem rather than of SA-FARI, the whole pipeline is re-run on a second dataset. Nothing changes but the data: the same frozen SAM 3, the same official scorer, the same distances, and the same leave-species-out cross-validation and paired group bootstrap.

This is cheap because the pipeline is bound to an annotation schema rather than to a loader. An offline adapter converts the new dataset into the SA-Co JSON that the loader and evaluator already read, so nothing downstream changes.

BURST. The replication runs on the animal subset of the BURST validation split (Section 2.7.2): 41 species across 132 video $\times$ class cells, scored by mask-HOTA on the native masks. Animal categories are selected by a WordNet filter [19], taking the hyponyms of `animal.n.01`, and mapped to the seven-level taxonomy wherever it resolves. It resolves in full for 25 of the 41 species, and the rest are left undefined. BURST supplies neither of the metadata axes SA-FARI does, so the cell key is the per-video sequence and leave-species-out is the only valid grouping. Both `pDetA` and `pAssA` are tested here, at the same bar as SA-FARI and with the same animal-size positive control (Section 3.5.3).

3.6.1 The controlled model swap

The replication varies the data and holds the tracker fixed. The swap does the reverse. Everything else stays constant — the same cells, the same ground truth, the same four distances, the same GLM under leave-species-out cross-validation — and SAM 3 is replaced by another open-vocabulary, text-promptable tracker. Only `pDetA` changes, because the distances are functions of the data and its ground truth rather than of the tracker. That is what isolates the tracker as the sole varying factor, and it is also why this is a swap rather than a replication.

Two challengers of deliberately different architecture are used. GLEE [31] is a single unified detect–segment–track model. Florence-2 [32] paired with SAM 2 [25] is a caption-style box generator, carrying no confidence score, followed by a mask-propagating tracker.

Two safeguards keep the swap fair. The first is a measurement gate. A challenger's per-cell `pDetA` must show real spread across cells before any regression is fitted on it. A regression on scores floored at zero has no variance to explain and would report a spurious null, so a challenger scoring ≈ 0 on a dataset is dropped there rather than modelled.

The second concerns scoring fairness. Open-vocabulary detectors calibrate their confidence to their own training domain, so a near-zero score may reflect the evaluator's confidence gate rather than a real detection failure. Every such score is therefore re-checked under a box-based metric, and again under a threshold-free average-precision metric with no gate at all. This gate is also what decides which dataset can host a fair swap, and it is why the swap runs on BURST rather than SA-FARI; Chapter 4 gives the measurements behind that choice. Each challenger configuration is fixed in advance from the model's own defaults, run once, and reported as-is (Section 4.11).

3.7 Reproducibility and implementation

The constraints that protect validity are stated where they apply: the frozen tracker (Section 3.1.1), the official evaluator (Section 3.1.2), the frozen reference (Section 3.4), and the retained hard negatives (Section 3.2). What remains is how they are enforced.

Each is a configuration toggle rather than a branch in the code, so an ablation changes a config file and nothing else. The pipeline is covered by hermetic tests, among them the leakage check of Section 3.4 and the recovery of planted effects on synthetic data. Split definitions, random seeds and reference sets are all persisted. Every number reported in this dissertation comes from that pipeline, and both experiments reproduce end to end.

All inference is single-GPU and inference-only, since no weights are ever updated.

Chapter 4

Results and Evaluation

This chapter reports what the measurements show. It opens by checking that the frozen tracker reproduces the published benchmark, then works through the model in the order the decision rule of Section 3.1.3 demands: the fitted coefficients, how much variance the distances explain, and the out-of-sample test that settles the research question. Further sections cover the species hold-out, false positives on hard negatives, the robustness variants and the familiarity proxy. The last three take the analysis outside SA-FARI, to a second dataset and to two other trackers, and draw the comparison together.

4.1 Measurement

Before any modelling, the measurement itself has to be sound. Zero-shot SAM 3 on the SA-FARI test split reproduces the published benchmark: the dataset-level (micro) `VEval` score is ≈ 0.65 mask-`pHOTA`, matching the operating point reported with the dataset.

Table 4.1 reports a different and lower quantity. It gives the three mask metrics — detection, association, and their combination — as support-weighted means over the 346 positive cells, which is the form the model is fitted on. Detection, the primary target, sits at 0.527, and combined `pHOTA` at 0.533. The gap from 0.65 is not a discrepancy. The micro figure pools every frame in the dataset into a single score, whereas this one averages within each cell first and counts positive cells only (Section 3.3).

Table 4.1: Zero-shot SAM 3 on the SA-FARI test split — support-weighted mean over 346 scored cells (of 1447).

Metric (mask)	Value
pDetA	0.527
pAssA	0.546
pHOTA	0.533

4.2 Fitted coefficients

This section reports the first of the two criteria a distance must clear (Section 3.1.3): does its coefficient's confidence interval exclude zero? Table 4.2 answers that for the location split, Split B, over the same 346 positive cells as Table 4.1. The species hold-out is reported separately in Section 4.5.

Each row of the table is one factor, and the two columns give its coefficient and confidence interval for detection and for association. Three things make it readable.

The coefficients are standardised. Each is the change in the logit of the score for a one-standard-deviation rise in that factor. Magnitudes can therefore be compared down a column, even though the raw factors are in quite different units — tree steps, cosine gaps, a binary flag. A negative coefficient means the score falls as the factor rises, which is the direction H1 and H2 predict.

The intervals are not the model's own. A distance is constant within a species or a location, so the 346 cells behave like only a few tens of independent groups. These intervals resample whole species and whole locations instead (Section 3.5.2). They are wide deliberately, and that width is the real uncertainty rather than a defect of the data.

Only three of the four distances appear. Temporal gap is absent because it carries no variance to estimate from and the model drops it (Section 3.4.4). The last two rows are a scene covariate and the support control rather than distances.

Read that way, the table says one thing: no factor clears the bar. Every interval spans zero, for detection and association alike. Criterion (i) already fails, before the out-of-sample test of Section 4.4 is reached.

Two details deserve reading rather than skipping. First, the point estimates for taxonomic and visual distance are negative here, nominally the direction H1 predicts. But the intervals are far too wide for that sign to carry information. And on the species hold-out, where species novelty actually varies, both coefficients come back positive (Section 4.5) — the opposite of what H1 says.

Second, taxonomic distance carries by far the widest interval, $[-3.87, +0.09]$ for detection. That is a property of this split rather than a finding about taxonomy. The official split shares species across train and test, so taxonomic distance is ≈ 0 for almost every cell here, and the coefficient is estimated from almost no variation. It is measured imprecisely, not measured and refuted, and we count it as neither. Testing species novelty properly is what the constructed species hold-out exists for.

Table 4.2: Standardised logit-GLM coefficients (group-cluster-bootstrap 95% CIs) for detection and association. Every distance's CI spans zero.

Factor	pDetA coef. [CI]	pAssA coef. [CI]
Taxonomic distance	-0.362 [-3.87, 0.09]	-0.368 [-3.40, 0.08]
Visual distance	-0.144 [-0.50, 0.14]	-0.122 [-0.49, 0.18]
Environment distance	0.042 [-0.19, 0.24]	0.013 [-0.23, 0.21]
Clutter	-0.066 [-0.38, 0.29]	-0.087 [-0.41, 0.28]
Night / IR	-0.405 [-0.94, 0.11]	-0.462 [-1.02, 0.11]
log(support)	-0.216 [-0.51, 0.17]	-0.224 [-0.53, 0.18]

4.3 How much the distances explain, and the difficulty pivot

The first question is how much of the variation in detection the four distances account for. The answer is almost none. A dominance analysis puts the four together at only $\approx 3\%$ of the variation, and no single distance stands out (Table 4.3). The method is LMG/Shapley, which splits shared explanatory power fairly between correlated predictors rather than double-counting it.

A natural worry is that the distances are so correlated with one another that a real effect cancels out between them. The VIF column rules that out. The variance inflation factor measures how entangled each predictor is with the others, and a value near 1 means essentially independent. Every VIF here is ≈ 1, so the 3% is genuinely all there is, not a masked signal.

The one distance that looked predictive was visual distance, and it turned out to be animal size in disguise. Visually distinctive species tend to be larger, and larger animals are easier to

Table 4.3: Variance partition for detection: each distance's unique share of R^2 (LMG/Shapley) and its VIF. The four distances explain only $\approx 3\%$ of the variance and every VIF ≈ 1 — the null is not masked by collinearity.

Factor	unique R^2	share	VIF
Taxonomic	0.014	48%	1.06
Visual	0.012	39%	1.06
Environment	0.004	13%	1.00
Temporal	0.000	0%	–

detect. What looked like novelty was a confound.

That prompts a different question. Rather than how novel the animal is, does how hard the scene is predict detection? Three label-free scene properties are tested against the same bar: low light, visual clutter, and a night/infrared flag (Table 4.4). Each is measured directly from the frames, with no SAM 3 run and no target label.

Bundled together with animal size, these difficulty signals appear to work, beating the baseline by $\approx +0.017$ on both hold-outs. Separating the contributions shows that the entire gain comes from size. Size *on its own* carries it. Low light *on its own* does not beat the baseline, and adds almost nothing once size is accounted for.

The apparent night/IR effect is inflated by being measured at the wrong level. Per location, its correlation with detection looks moderate, at $r = -0.377$. That is an averaging artefact. Measured at the level of the individual cell, where the model actually operates, it shrinks to -0.13, and clutter's is 0.00. So the difficulty idea is **also null**: the only label-free property that predicts detection is animal size.

Table 4.4: Nested size decomposition of the intrinsic-difficulty test (detection, 346 cells). Entries are the out-of-sample MAE gain over a mean baseline, Δ [95% CI] (p), under leave-species-out and leave-location-out. The entire gain sits in the size covariate; low-light *alone* does not validate — the difficulty signals pass only by riding on size.

Model	Leave-species-out Δ [CI] (p)	Leave-location-out Δ [CI] (p)
log(support) only	+0.001 [-0.007, +0.008] (0.432)	+0.002 [-0.005, +0.008] (0.291)
log(area) only (size)	+0.015 [+0.002, +0.028] (0.014)	+0.017 [+0.009, +0.025] (0.001)
Low-light only (no size)	+0.003 [-0.006, +0.011] (0.233)	+0.004 [-0.003, +0.011] (0.123)
Low-light + size	+0.017 [+0.004, +0.032] (0.011)	+0.020 [+0.011, +0.030] (0.000)

4.4 The headline test: predicting unseen species and locations

This is the decisive test. The model is fitted on some species and locations, then asked to predict detection for held-out ones it never saw. Its error is compared against the simplest possible guess: always predicting the average score.

Table 4.5 reports that comparison. Each row is one combination of hold-out scheme and target, so the four rows cover leave-species-out and leave-location-out for both detection and association, all within the same 346 positive cells. The MAE column is the model's mean absolute error, and Baseline is the mean predictor's. Δ is the gain of one over the other, so a positive value means the model beats guessing the mean. The interval comes from the paired group-bootstrap, and the p-value is one-sided (Section 3.5.3).

The result is a **null**. Every gain falls between +0.004 and +0.007, every interval spans zero, and no p-value drops below 0.13. On both hold-out schemes, and for both targets, the distance model does no better than the trivial average.

Figure 4.1 shows what that looks like. Each point is one held-out cell, with its true detection score on the horizontal axis and the model's out-of-sample prediction on the vertical. A model that predicted well would place its points along the dashed diagonal. A model that ignored its inputs and returned the average every time would place them along the horizontal line. The points follow the horizontal line: whatever the truth, the prediction barely moves.

The null holds up because the test demonstrably works. Run unchanged on the same cells, it *does* recover a real effect. Animal size beats the baseline out of sample at the identical bar, significantly so on both schemes (ΔMAE = +0.015 leave-species-out, +0.017 leave-location-out; Table 4.4). A test that fires for size while staying flat for the distances tells us the distances genuinely carry *no* predictive signal. The flat result is an absence of signal, and the size result is what confirms it.

Table 4.5: Out-of-sample error vs a mean predictor on the **location split** (Split B; the 346 positive cells). Each row is one cross-validation grouping scheme — leave-species-out or leave-location-out — applied within these same cells; Δ is the MAE gain over baseline, with a paired-bootstrap 95% CI and one-sided p. No scheme's gain is significant.

CV scheme	Target	n	MAE	Baseline	Δ [95% CI]	p
location	pAssA	346	0.289	0.295	+0.007 [-0.005, +0.018]	0.13
location	pDetA	346	0.293	0.298	+0.004 [-0.007, +0.015]	0.23
species	pAssA	346	0.289	0.295	+0.006 [-0.006, +0.017]	0.16
species	pDetA	346	0.293	0.298	+0.004 [-0.006, +0.015]	0.24

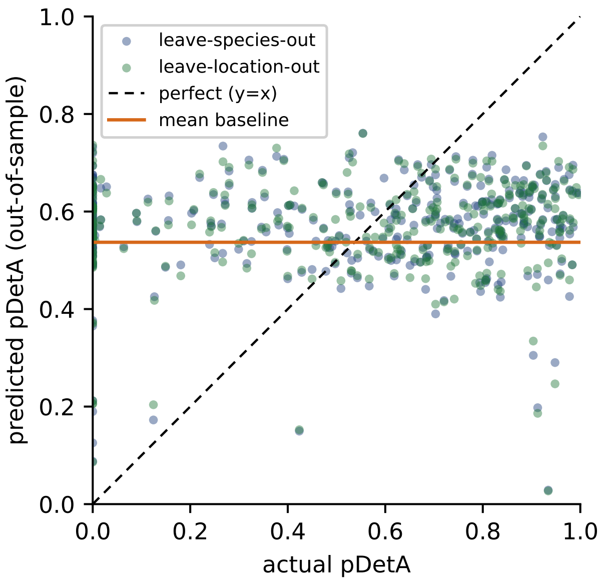

Figure 4.1: Out-of-sample detection predictions against the truth, for held-out species (navy) and held-out locations (green); each point is one cell. Perfect prediction would lie on the dashed diagonal, and predicting the mean regardless of input lies on the orange horizontal. The points follow the orange line, so the distances carry no predictive power.

4.5 Species hold-out (H1): novelty and the size confound

The primary novelty test is a leave-one-species-out analysis over the pooled train and test species: 99 species and 1025 scored cells. Three tables report it, each the Split A counterpart of one already seen for Split B. Table B.1 gives the measurement, and confirms the tracker scores in the same range on the pooled set as on the test split alone. Table 4.6 gives the fitted coefficients, and Table B.3 the out-of-sample test.

Out-of-sample prediction fails again. Every gain in Table B.3 is zero or negative, so the distance model is no better than guessing the mean, and on the species scheme slightly worse.

Table 4.6 is where this split differs from Split B, and it repays careful reading. One coefficient does have an interval excluding zero: visual distance, at +0.37 for detection. Its sign, though, is *positive*. Read literally, that says more-novel species are detected better, which is the reverse of what H1 predicts.

It is a **confound** rather than a reversal. Visually distinctive species tend to be larger (visual $\leftrightarrow$ log-area $r = 0.22$), and larger animals score higher (log-area $\leftrightarrow$ `pDetA` $r = 0.30$, the strongest raw correlate of the score). Adding an animal-size covariate shrinks the visual coefficient from +0.37 to +0.22, and its interval now spans zero, while size itself becomes the significant driver. "Visual novelty helps" was really "bigger animals are easier". Taxonomic novelty, meanwhile, is flat ($r \approx 0.00$).

Figure 4.2 shows that underlying relationship directly. Each point is one cell, with animal size on the horizontal axis and its detection score on the vertical, and the fitted trend rises at $r = +0.31$. Novelty does not predict transfer. Size does, a little.

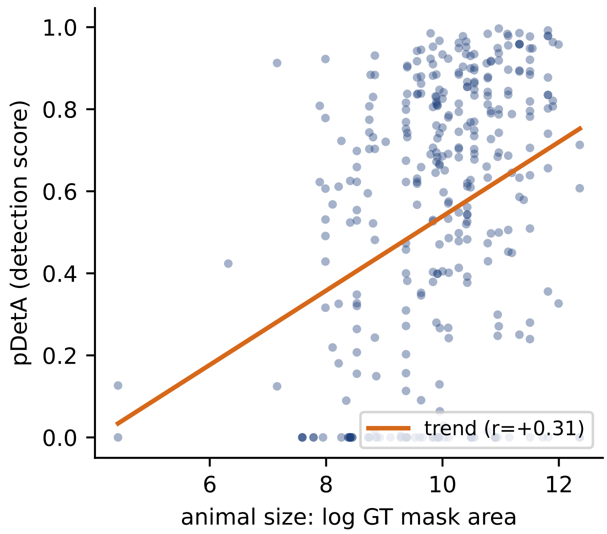

Figure 4.2: The one label-free property that predicts detection: animal size. Per-cell detection score against log ground-truth mask area, on the location split (346 cells), with the fitted trend at $r = +0.31$. This is the confound behind the apparent visual-distance effect.

Table 4.6: Standardised logit-GLM coefficients (group-cluster-bootstrap 95% CIs) for detection and association, species hold-out.

Factor	pDetA coef. [CI]	pAssA coef. [CI]
Taxonomic distance	0.202 [-0.09, 0.43]	0.209 [-0.06, 0.42]
Visual distance	0.369 [0.11, 0.67]	0.271 [0.04, 0.54]
Environment distance	0.043 [-0.15, 0.17]	0.023 [-0.15, 0.13]
Clutter	-0.284 [-0.52, 0.02]	-0.230 [-0.46, 0.06]
Night / IR	-0.286 [-0.77, 0.22]	-0.439 [-0.92, 0.06]
log(support)	-0.020 [-0.14, 0.16]	0.051 [-0.06, 0.23]

4.6 False positives / hallucination

Everything so far has concerned misses: whether the tracker finds the animals that are there. The detection score conditions on the animal being present, so it measures recall and says nothing about hallucination. The opposite failure — reporting an animal that is not there — is measured on the 1486 hard negatives, the queries naming a species absent from the clip. A correct run returns nothing. SAM 3 returns a masklet anyway about 9.8% of the time.

Table 4.7 asks whether that rate is predictable from the same label-free distances. Each row is one species-level test, giving the statistic, its 95% confidence interval, and the verdict. The first four rows are correlations between a species' hallucination rate and one property. The last row is the out-of-sample test, where the statistic is an MAE gain over a mean predictor rather than a correlation.

This is the closest the study comes to a validated before-running signal. Taxonomic novelty does nothing ($r \approx +0.06$, not significant). Visual distinctiveness does: the more distinctive a species looks, the *less* the model hallucinates it ($r \approx -0.33$). That survives an animal-size control, strengthening to a partial -0.37, and size on its own is not significant. Unlike every earlier effect, this one is not size in disguise.

It remains a correlate rather than a predictor. Predicting a held-out species' hallucination rate from visual distance beats a mean baseline by only +0.008 MAE, and just misses out-of-sample significance at $p = 0.053$. Even the study's best signal does not clear the bar. Its direction is the opposite of the hypothesis too: distinctive means safer, rather than novel meaning worse.

Table 4.7: False positives on the 1486 hard negatives (113 species): overall hallucination rate 9.8%. More visually distinctive species hallucinate *less* — a real, size-independent correlate — but it does not validate out of sample (leave-species-out $p = 0.053$): a correlate, not a predictor.

Test (species-level)	statistic	95% CI	verdict
FP rate vs taxonomic distance	+0.06	[−0.17, +0.23]	n.s.
FP rate vs **visual distance**	−0.33	[−0.43, −0.23]	significant
visual, size-controlled (partial)	−0.37	[−0.47, −0.28]	significant
FP rate vs animal size	+0.13	[−0.06, +0.31]	n.s.
leave-species-out OOS gain	+0.008	[−0.002, +0.017]	$p = 0.053$

4.7 Ablations and robustness

The null of Section 4.4 rests on one particular model, built from one particular set of design choices. This section asks whether either could be responsible for it. The first subsection varies which factors enter the model. The second holds the factors fixed and varies how they are computed.

4.7.1 Factor ablation

Table B.4 refits detection with one change to the design per row. The pseudo-R^2 column is McFadden's, a measure of in-sample fit. The last two columns give the out-of-sample MAE gain over a mean baseline under the two hold-out schemes, leave-species-out and leave-location-out, each with its p-value.

Dropping any distance in turn barely moves either quantity, so no single distance is carrying signal that the others obscure. Dropping temporal gap changes nothing at all, which is expected: it has no variance and the model already discards it (Section 3.4.4). Removing the support covariate leaves the model slightly worse than the baseline (-0.002 and -0.003), so the null is not that control suppressing a real effect.

One row does move. Adding the animal-size covariate more than doubles the fit, from 0.055 to 0.119, and lifts the out-of-sample gain to significance on both schemes (+0.018 at $p = 0.02$, +0.020 at $p = 0.00$). That is the conclusion the variance partition reached from the other direction (Table 4.3): size is the only real correlate.

4.7.2 Design robustness

Table B.5 keeps the factor set fixed and swaps a single design choice per row. Each entry is the out-of-sample gain with its confidence interval and p-value, under both hold-out schemes, judged after a Bonferroni correction over the five variants.

Three swaps test the distances themselves: changing the encoder from DINOv2 to CLIP, taking a whole-frame embedding that keeps the background instead of a mask crop, and replacing the nearest-prototype distance with a distributional Fréchet or MMD distance. All leave the gain indistinguishable from the baseline. Every ΔMAE is at most 0.007, every p-value at least 0.10, and every interval spans zero.

The generic "animal" prompt is included for completeness, and behaves differently. Prompting non-specifically collapses per-species detection to a near-constant, mostly-zero target: mean `pDetA`$\approx$ 0.14, with 73% of cells exactly zero. Fitting a target that degenerate, the model does not merely fail to beat the mean but falls far below it ($\Delta\text{MAE} \approx -0.35$).

Across every variant the label-free distances fail identically. The null is a property of the transfer problem, not of one modelling choice.

4.8 SAM 3 familiarity proxy

Every distance so far has measured novelty against the SA-FARI reference. That invites an objection: SA-FARI is not what SAM 3 was trained on, and its real pretraining set is undisclosed. Perhaps the distances fail only because they measure novelty from the wrong anchor.

The familiarity proxy (Section 3.4.6) tests that objection by reading familiarity from SAM 3 itself, from how separable each species is in its own vision-encoder feature space. Table B.6 reports the three separability metrics over the 346 positive cells. The first two columns give each metric's out-of-sample gain and p-value, on SA-FARI and on BURST. The next two give its correlation with the DINOv2 visual distance and with animal size, which together show what the metric is really measuring. The last column is the verdict against a Bonferroni threshold of $\alpha = 0.017$ for three metrics.

The answer is another null. None of the three clears the bar on the leave-species-out novelty test. The strongest, silhouette, reaches only $p = 0.034$ for a gain of +0.012, and nearest-prototype and Mahalanobis do not come close, at $p = 0.068$ and 0.085. The null replicates on BURST, where the same three give $p = 0.020$, 0.136 and 0.148.

The r_{vis} column explains why. All three correlate with the DINOv2 visual distance they re-encode, at 0.63 to 0.85. They largely restate a feature already shown to be null, and in a head-to-head none adds out-of-sample gain over the visual distance plus size. One row is a genuine curiosity: silhouette, unlike the visual distance, is *not* a size artefact ($r_{\text{size}} = -0.16$). It isolates the size-independent half of visual distinctness, and still fails to predict transfer.

The metrics do reach significance under leave-location-out ($p \leq 0.01$), but that is not a novelty test. A familiarity value is constant within a species, with zero within-species variance. On the location split the held-out locations' species also appear in training, so a species-constant feature is never tested on an unseen species at all.

Nor is the null a power failure. On these same 346 cells the animal-size control does validate out of sample (Table B.4), so a test that stays flat here is finding nothing rather than lacking the sensitivity to find it. Reading familiarity from SAM 3's own representations therefore does not rescue before-running prediction. It recovers the same size-entangled visual distinctness that was already null, and it closes the "but you do not know SAM 3's true training set" objection to that null.

4.9 Independent replication on a second dataset

Both SA-FARI experiments overlay different splits on one frozen inference pass, so they are near-orthogonal rather than independent (Section 3.6). BURST [1] supplies the independent test: 132 cells across 41 species, mask-native, in a quite different regime of internet, driving and film video rather than camera traps.

Two preconditions hold there. Zero-shot SAM 3 is non-degenerate, at pHOTA 0.635 (Table B.2), so there is real variance to model. And the association target is genuine rather than near-degenerate as it is on SA-FARI: `pDetA` and `pAssA` correlate at 0.76, and of the 71 multi-object cells not one has them exactly equal. Both components are therefore testable.

Table B.8 reports the outcome under leave-species-out cross-validation. Each row is one predictor, with its out-of-sample MAE gain, confidence interval and p-value. The last row is the animal-size control rather than a distance.

The distances fail again. Combined, they improve out-of-sample error by only $+0.002$ ($p = 0.374$), and individually visual ($+0.003$), taxonomic (-0.006) and environment (-0.001) are all null. So are the association specifications. The result survives dropping the 19 single-cell species, and survives a size control.

The one near-signal is the familiar artefact. Visual distance correlates with detection at $+0.17$ per cell, and at $+0.37$ between species — the same aggregation inflation seen for night/IR in Section 4.3. Both are *positive*, meaning more-novel animals are detected better, which is the opposite of the novelty hypothesis. And the correlation dissolves under a size control (partial $+0.08$, $p = 0.35$).

This is a second convergent null rather than positive proof. No before-running quantity reaches significance here, not even the animal-size control that does validate on SA-FARI. Without a control that fires, the test cannot certify that it would have caught a small species-level effect. What BURST establishes is that no moderate-or-large before-running distance signal exists in an independent mask-native regime with a genuine association target. A small species-constant effect cannot be excluded.

4.10 Cross-dataset synthesis

Table 4.8 collects every feature tested, its result on each dataset, and the verdict. The two dataset columns are not in the same metric, and each header says which it is. BURST entries are out-of-sample ΔMAE with a p-value. SA-FARI entries are standardised GLM coefficients with confidence intervals, except the two rows marked †, which are ΔMAE like BURST. A feature predicts transfer only if it is positive *and* significant.

The pattern is consistent. No before-running label-free distance, alone or combined, predicts detection or association transfer out of sample on either dataset, and every gain over a mean predictor sits within bootstrap noise of zero.

The only near-signal is visual distance, and it points the wrong way: more-novel animals are detected better, not worse. It dissolves under an animal-size control wherever it is examined.

One row does clear the bar, and it is the confound rather than a hypothesis variable. Animal size validates out of sample on SA-FARI, modestly. That row doubles as the positive control: the same test that stays flat for every distance does detect size, so the null is an absence of signal rather than an absence of power.

The contribution is that null, with its bounds stated. It is consistent across a camera-trap dataset and an internet-video one, and powered on SA-FARI to exclude a moderate effect, though not a small one.

Table 4.8: Cross-dataset synthesis: every feature tested, its value on each dataset, and the verdict. Each column is reported in one metric, stated in its header: on **BURST** every entry is the out-of-sample ΔMAE (p) over a mean predictor; on **SA-FARI** the individual distances are the standardised GLM coefficient with its 95% CI, while the two rows marked † (combined distances, animal size) are ΔMAE (p) like BURST. A feature "predicts transfer" only if it is positive *and* significant. Only the animal-size confound clears the bar, and only on SA-FARI (the positive control); every label-free distance is null on both datasets.

Feature	SA-FARI (coef. [CI])	BURST (ΔMAE, p)	Verdict
Before-running label-free distances (the research question)			
Combined distances†	-0.004 (0.74)	$+0.002$ (0.37)	null on both
Visual alone	-0.14 $[-0.50, 0.14]$	$+0.003$ (0.26)	null; wrong direction
Taxonomic alone	-0.36 $[-3.87, 0.09]$	-0.006 (0.95)	null on both
Environment alone	$+0.04$ $[-0.19, 0.24]$	-0.001 (0.75)	null on both
Temporal gap	32/1447, all $= 0$	no timestamps	untested
Confound control			
Animal size (log-area)†	$+0.015$ (0.01)	$+0.005$ (0.14)	**validates** (SA-FARI)
Detection precision (hard negatives)			
Hallucination vs. visual	$r = -0.33$; $+0.008$ (0.053)	—	marginal correlate

4.11 Cross-model replication

Both datasets above vary the data while holding the tracker fixed, which leaves one threat open: the null could be a quirk of SAM 3 rather than a property of the prediction problem. The controlled swap of Section 3.6.1 closes it by changing only the tracker, using two challengers of deliberately different architecture, GLEE and Florence-2 paired with SAM 2.

Why the swap runs on BURST. On SA-FARI both challengers score pDetA ≈ 0, leaving no variance for a regression to explain. That is a genuine detection failure rather than an artefact of the evaluator. The zeros survive box-HOTA, which scores the box models on boxes (GLEE 0.033), and they survive a threshold-free average-precision metric with no confidence gate at all (GLEE 0.000, Florence-2 0.017–0.022). Florence-2 emits no confidence score in the first place, so no gate can be blamed, and it still scores near zero.

The cause is the domain rather than the species. Run GLEE on SA-FARI's everyday animals — jaguar, cow, fox — and it localises them well, at oracle IoU up to 0.90. Its confidence still collapses below any usable operating point on the dark, cluttered camera-trap footage. SA-FARI is also SAM 3's own co-released benchmark. Neither condition holds on BURST, where SAM 3 in fact scores higher (pDetA 0.607) than on SA-FARI (0.537), so there is no home-turf advantage to correct for.

Both challengers track. Table 4.9 gives each tracker's mean detection score over BURST's 132 cells. GLEE reaches 0.340 and Florence-2 + SAM 2 reaches 0.394, against SAM 3's 0.607. Both sit well clear of zero, and more importantly both show real per-cell spread: medians of 0.31 and 0.33, with 67 and 72 of the 132 cells above 0.3. That is the variance the measurement gate requires.

Table 4.9: Cross-model swap on BURST (132 cells, 41 species, mask-HOTA): zero-shot detection per tracker — SAM 3 and two architecturally distinct challengers. All three are non-degenerate: both challengers show the real per-cell spread the regression needs (GLEE median 0.31, 67/132 cells > 0.3; Florence-2 + SAM 2 median 0.33, 72/132 cells > 0.3). Their distance-model results are in Table B.7.

Tracker	mean `pDetA`
SAM 3 (reference)	0.607
GLEE (unified query)	0.340
Florence-2 + SAM 2 (caption)	0.394

The null replicates. Table B.7 applies the same leave-species-out, size-controlled procedure to each challenger. The combined four-distance model predicts neither: +0.011 for GLEE (95% CI $[-0.004, +0.027]$, $p = 0.088$) and +0.034 for Florence-2 ($[-0.001, +0.068]$, $p = 0.030$). Both intervals span zero, and neither clears the Bonferroni bar of $\alpha = 0.0125$.

Individual distances flicker, as they did on SA-FARI — environment reaches $p = 0.006$ for Florence-2 — but the same discounts apply. BURST has no locations, so leave-species-out is the only valid partition, and an apparent pass on two schemes is one partition counted twice. The flickers route through the animal-size confound that the driver controls for. And the combined model, which is what a deployment gate would actually use, is null for both.

Two architecturally distinct trackers therefore both track BURST's animals and both show the before-running distance null. **The null is task-general across trackers, not a SAM-3 artefact.**

Convergent, but bounded. This is a controlled swap rather than an independent replication: it reuses BURST's cells, ground truth, distances and GLM, changing only the tracker. Its scope is two models on one dataset of 132 cells, underpowered for a small species-level effect exactly as the SAM 3 run there was, and Florence-2's interval sits closer to excluding zero than GLEE's. We therefore read it as a convergent, confound-free cross-model replication that removes the single-backbone threat, not as an independently powered certification. Combined with the two datasets, the null now holds across both axes that could have confounded it: the dataset and the model.

Chapter 5

Discussion

Chapter 4 reported what the measurements show. This chapter asks what they mean. It sets out the interpretation the evidence supports, answers the objection that novelty must surely matter, reconciles the null with a literature of positive results, draws out what follows for practice, and states the bounds within which the claim holds.

5.1 What the result means

Neither hypothesis survives. H1 and H2 both fail the bar set in advance in Section 3.1.3, on both hold-out schemes and for both targets (Section 4.4). H0 is what the data support: transfer is not distance-driven.

How they fail matters as much as that they fail. The one coefficient that reached significance ran in the wrong direction, and dissolved under a size control. The follow-up difficulty pivot behaved the same way, its apparent effect routing entirely through animal size. So the only label-free property that predicts detection is the very quantity the analysis introduced as a nuisance to control for.

The detection–association decomposition, the intellectual core of the design, produced a second and quieter finding. Association is a near-degenerate target on this data. Of the 346 positive cells, 268 (77%) hold at most one annotated animal per video, so there is only one identity to keep. Across all positive cells `pAssA` correlates with `pDetA` at 0.97 and is exactly equal to it in 73%, leaving little identity-specific variance for any feature to predict. The decomposition is therefore a detection story, and the association null is reported with those divergence figures as its explanation rather than as a separate failed search.

5.1.1 Novelty at the extremes versus graded prediction

A natural objection is that novelty surely matters. A familiar species is detected better than a totally alien one, so how can a novelty distance be null?

Both statements are true, because they concern different resolutions of the same axis. The coarse claim, that the extreme tails differ, is real and never in dispute. The null reported here is the far stronger graded claim that H1 actually tests: that a continuous distance predicts where on the detection scale an unseen species lands, out of sample, beyond the size confound, and well enough to gate a deployment.

The two come apart because the informative variance lives in the middle of the range rather than at the poles. The easy extremes are already predicted by the mean, with common large animals near 1 and hopeless cases near 0. A coarse alien-versus-familiar signal therefore adds nothing exactly where prediction would have value. A genuine extreme-tail effect can coexist with a null graded predictor without contradiction, and it is the graded, held-out, size-controlled predictor that a practitioner would need.

5.2 Why the null does not contradict prior positives

The result sits against two literatures of positive findings: one that predicts model performance without labels, and one that shows backgrounds drive camera-trap failure. Neither is contradicted here, and it is worth saying precisely why.

5.2.1 Label-free performance estimation

Two axes separate this study from every positive result in Section 2.4.

The first is the signal. Those estimators all read the model's own behaviour on the target, after running it. This work forecasts from external distances known before a single frame is processed. That is the strictly harder regime, and for a practitioner deciding whether to spend compute, the more useful one.

The second is the output. Every predecessor targets a scalar classifier accuracy, or the quality of a static detector or segmenter. None concerns identity maintained over time, and none decomposes its estimate. That is the gap identified in Section 2.8.

AutoEval [10] is the closest methodological cousin, and the strength of its positive sharpens the contrast. It reports a near-perfect rank correlation between a Fréchet distribution-shift distance and classifier accuracy. That distance, though, is computed from the model's own features after the target images have passed through the network. We tested AutoEval's exact functional in our own regime — Fréchet and MMD distances on cached DINOv2 embeddings, computed before running SAM 3 — and it stayed null (Section 4.7). The failure is therefore not an artefact of a lazy nearest-prototype distance. Our nearest analogue does worse still, running in the opposite direction before dissolving under a size control.

One bridge makes the boundary concrete. Accuracy-on-the-Line [20] was validated on iWildCam, the same camera-trap shift this dissertation cites through WILDS and Terra Incognita [3, 16]. The exact wildlife shift where a classifier's out-of-distribution accuracy is predictable label-free is where our before-running distances are not.

5.2.2 Background bias

The environment axis inherits its foreground/background separation from a literature that finds backgrounds emphatically do matter (Section 2.5), so its null needs reconciling.

Two distinctions dissolve the tension. The first is causal reliance versus predictive distance. Xiao et al. [33], Rosenfeld et al. [26] and PanAf-FGBG [5] all intervene on the background and watch performance move. That is a statement about how a model uses context. Ours is the strictly weaker, correlational question of whether a scalar scene distance forecasts an untouched model's per-cell score. A model can lean causally on context while its score stays unpredictable from a one-number distance. The null does not deny their causal findings. It locates a regime where the distance shortcut fails.

The second distinction is why the promptable regime should be that regime, and the same literature supplies the mechanism. Those models must infer the category, and a spuriously correlated background is a usable shortcut for doing so. SAM 3 is told what to find. The prompt `"impala"` supplies the foreground identity and removes the incentive to ride the scene — exactly the direction Xiao et al. [33]'s own finding predicts, since better and less shortcut-prone models close the background gap. PanAf's target compounds the difference, since fine-grained behaviour is intrinsically background-correlated whereas finding and following a named animal is not.

The result is therefore an extension rather than a refutation. The backgrounds that causally drive a camera-trap classifier's location drop do not, once isolated from the animal and used as a before-running distance, forecast a promptable tracker's transfer. Because we isolate scene from animal explicitly, a step none of the label-free-prediction methods take, this is a cleaner statement than "background is diagnostic" rather than a weaker one. The claim is bounded accordingly: it concerns background distance as a predictor of detection transfer for a promptable tracker, not a blanket claim that background is irrelevant.

5.3 What this means in practice

The field problem that motivated this work — telling a practitioner, before spending compute, whether a promptable tracker is trustworthy for a new species and place — is not solved by label-free distances. That is a negative answer, but a useful one. The intuition that far-from-training implies unreliable does not hold here, so a deployment gate cannot be built from taxonomic, visual, environmental or temporal distance to a reference set. The one predictive label-free signal, animal size, is too coarse and too obvious to serve as one.

One after-running signal does clear that bar, and it is stated plainly here rather than left implicit. If a practitioner is willing to run the tracker first, an ATC-style estimator [13] — reading SAM 3's own detection confidence on the target and thresholding it — predicts detection out of sample on both cross-validation schemes, where every label-free distance failed. It is deliberately kept outside the contribution, for three reasons. It is after-running, a weaker operating point than the before-running question this work set out to answer. It is confirmatory of an established family rather than novel. And it is mechanistically close to circular, since feature and target are both functions of SAM 3's own output: a cell with no confident masklet trivially has zero detection accuracy. It is a real, validated estimator for the after-running, before-labelling operating point — but not the before-running predictor this dissertation asks for, and not part of the null it reports.

5.4 Limitations and threats to validity

The reference is not the pretraining corpus. Distances are measured against the SAFARI train split, not against SAM 3's true pretraining data, which is undisclosed. The null is scoped accordingly. One hedge was tested: the familiarity proxy reads separability directly from SAM 3's own features (Section 4.8), and it too is null, recovering the same size-entangled visual signal. Even model-internal familiarity does not lift the result. A coverage-anchored proxy for the actual corpus remains untested.

One dataset and one backbone. A single inference pass is reused across both splits, so the species and location experiments are two overlays on the same scored cells rather than independent replications. Two steps were taken against this. Against the dataset threat, the whole pipeline was re-run on an independent dataset, where the null holds again (Section 4.9) — a second convergent null rather than positive proof, since that dataset remains underpowered for a small species-constant effect. Against the model threat, the controlled swap put two architecturally distinct trackers through the same procedure with the same result (Section 4.11).

A third dataset, MammAlps [12], was run in an earlier pass. It is directionally consistent with the null but is not counted here: at roughly 20 cells over 5 species it is too small to certify one, and its per-clip annotations are not reproducible from the public release. It is reported as excluded rather than silently dropped.

The effective sample is groups, not cells. Tens of independent species and locations, not hundreds of cells. That is why inference is group-clustered throughout, and why aggregate correlations are reported alongside their much weaker cell-level values, to avoid the ecological-correlation fallacy.

Representational probing is only partly opened. The familiarity proxy establishes that SAM 3's features separate the species, but not what those features encode. Section 6.3 takes this up.

The scorer. All scores come from the unmodified official evaluator, which wraps the standard HOTA-family metrics rather than any SAM-3-specific logic. Its one SAM-3-calibrated knob is a fixed 0.5 confidence gate, and Section 4.11 establishes that this gate is not load-bearing for the cross-model comparison. The measurement is therefore model-fair.

5.5 Ethical considerations

The data are anonymised camera-trap footage released under a non-commercial licence, and location identifiers are already anonymised in SA-FARI. The negative result carries a responsible-deployment message. Because label-free distance does not reliably flag when a promptable tracker will fail, conservation teams should not treat low distance from training data as a safety guarantee. Reliability should be checked with labelled spot-audits rather than assumed.

Chapter 6

Conclusion

6.1 Summary of findings

Camera traps and laboratory rigs now record wildlife video far faster than anyone can annotate it, and that annotation step is what limits how much ecology and behavioural science actually gets done. Promptable trackers offer a way past it. Name an animal in ordinary words, and the model finds and follows every one that appears, without ever having been trained on that species. The difficulty is that the ability is uneven, and uneven in ways nobody can currently anticipate. A team pointing such a model at a new species in a new place has no principled basis for deciding whether to trust what comes back, which leaves two unattractive options: trust it blindly, or fall back on the manual annotation the model was meant to replace.

This dissertation asked whether that judgement can be made in advance. Can a promptable tracker's zero-shot accuracy on an unseen species and place be predicted before the model is run, from properties that require no labelling of the target? It asked the question separately for the two halves of the task — finding the animal, and following it — because the two can fail for different reasons.

The answer is no, and it survives every criterion applied to it. No distance's coefficient interval excludes zero. None beats a mean predictor out of sample, on either a species or a location hold-out. Together the four explain about 3% of the variation in detection. The null holds when the encoder, the crop, the prompt and the distance functional are each swapped in turn. It holds for a second family of label-free properties, intrinsic scene difficulty, whose apparent effect collapses into animal size. It holds when familiarity is read from the tracker's own feature space instead of from an external reference. It holds on a second, independent dataset, and on two architecturally distinct trackers substituted for SAM 3. And it is not a failure of power: the same procedure detects animal size out of sample wherever size is present.

Only that one property predicts detection, and only modestly — the nuisance the analysis introduced in order to control for it. Association carries almost no variance distinct from detection on this data, so the decomposition that motivated the design turned out to be a detection story. What remains is a rigorous, decomposed, cross-dataset and cross-model null.

6.2 Contributions

To our knowledge this is the first study to test, out of sample, whether a promptable video tracker's zero-shot transfer is predictable from label-free signals known before the model is run, and the first to decompose that question into detection and association.

What the field gains is threefold. The formulation and the protocol are reusable: any future before-running estimator can be held to the same whole-group hold-out bar that these well-motivated distances failed, which is what makes a negative answer useful rather than merely discouraging. The explanation is specific rather than vague: the apparent signals reduce to an animal-size confound, so any future study measuring visual novelty across species must control for apparent size or it will rediscover the same artefact. And the caution is practical: proximity to familiar training data is not a safety guarantee, so reliability has to be checked rather than assumed.

6.3 Future work

Three directions remain open.

The first is a fuller representational probe. The familiarity proxy tests only whether SAM 3's features separate the species, and finds that they do by re-encoding the visual distance (Section 4.8). What those features actually encode — whether taxonomy is represented structurally — is untested, as is a proxy anchored on the coverage of the true, undisclosed pretraining corpus.

The second is a broader model swap: more trackers, and ideally a larger fair dataset that is no tracker's home turf. SA-FARI's larger cell count could not be reused here, for the reason given in Section 4.11.

The third is targets with richer association structure. In crowded, multi-object footage the detection–association decomposition would have considerably more to distinguish than it did here.

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Appendix A

Mathematical definitions

Chapter 3 defines the tracking metric, the four distances and the regression in the body. This appendix supplies only what was left out of it: the scene covariates, the design matrix, and the two statistics used to report fit and out-of-sample gain.

A.1 Scene covariates

Three interpretable covariates accompany the environment distance (Section 3.4). Writing f for a frame, ℓ for a location, and the bar for a mean over that location's frames,

$$\text{achromatic}(f) = \Big[\, \overline{\max_c f_c - \min_c f_c} \;<\; \tau_{\text{ir}} \,\Big], \qquad \tau_{\text{ir}} = 12, \tag{A.1}$$

$$\text{is\_night\_ir}(\ell) = \big[\, \overline{\text{achromatic}(f)} > 0.5 \,\big], \qquad \text{clutter}(\ell) = \overline{\text{Var}\big(\nabla^2 f\big)}, \tag{A.2}$$

where $\max_c - \min_c$ is the per-pixel spread across colour channels and ∇^2 is the four-neighbour Laplacian on the greyscale background.

A.2 The design matrix

Equation 3.4 fits $\operatorname{logit} \mathbb{E}[y] = x^\top \beta$ with variance weights $w = n_{\text{frames}}$. The design x standardises each continuous predictor as $z = (x - \mu)/\sigma$, imputing missing values to the training mean μ and dropping zero-variance columns. It adds the standardised $\ln(1+n_{\text{frames}})$ as the support covariate, passes the binary is_night_ir through unchanged, and adds an intercept. Every standardisation statistic is learned on training rows only.

A.3 Fit and out-of-sample gain

Fit quality is the McFadden pseudo-R^2. The per-cell out-of-sample gain over a mean predictor is δ_i:

$$R^2_{\text{McF}} = 1 - \frac{D}{D_0}, \qquad \delta_i = \underbrace{|y_i - \bar{y}|}_{\text{baseline error}} - \underbrace{|y_i - \hat{y}_i|}_{\text{model error}}, \tag{A.3}$$

with D and D_0 the model and null (intercept-only) deviances. A positive δ_i means the distance model beats guessing the mean on cell i.

For the species-level false-positive analysis (Section 3.5.6), correlations are weighted by the per-species probe counts w:

$$r_w = \frac{\sum_i w_i (x_i - \bar{x}_w)(y_i - \bar{y}_w)}{\sqrt{\sum_i w_i (x_i - \bar{x}_w)^2}\,\sqrt{\sum_i w_i (y_i - \bar{y}_w)^2}}, \qquad \bar{x}_w = \frac{\sum_i w_i x_i}{\sum_i w_i}. \tag{A.4}$$

Appendix B

Supplementary tables

These tables support claims made in Chapter 4. Each is referenced from the section that discusses it.

Table B.1: Zero-shot SAM 3 on the pooled train+test set — support-weighted mean over 1025 scored cells (of 2126).

Metric (mask)	Value
pDetA	0.536
pAssA	0.597
pHOTA	0.559

Table B.2: Zero-shot SAM 3 on the BURST val animal subset — support-weighted mean over 132 probe cells (41 species), mask-HOTA. The score is non-degenerate and the association target is genuine (corr(`pDetA`, `pAssA`) = 0.76, 71/132 multi-object cells).

Metric (mask-HOTA)	Value
pDetA	0.607
pAssA	0.688
pHOTA	0.635

Table B.3: Out-of-sample error vs a mean predictor on the **species hold-out** (Split A; the 1025 pooled train+test cells over 99 species). Each row is one cross-validation grouping scheme — leave-species-out or leave-location-out — applied within these same cells; Δ is the MAE gain over baseline, with a paired-bootstrap 95% CI and one-sided p. No scheme's gain is significant. (Distinct experiment from Table 4.5, hence the different n and values.)

CV scheme	Target	n	MAE	Baseline	Δ [95% CI]	p
location	pAssA	1025	0.270	0.270	-0.000 [-0.006, +0.005]	0.52
location	pDetA	1025	0.274	0.274	-0.001 [-0.007, +0.006]	0.58
species	pAssA	1025	0.273	0.270	-0.003 [-0.014, +0.008]	0.74
species	pDetA	1025	0.277	0.274	-0.004 [-0.015, +0.009]	0.74

Table B.4: Factor ablation for detection: McFadden pseudo-R^2 and the out-of-sample MAE gain over a mean baseline under leave-species-out (LSO) and leave-location-out (LLO). Dropping any distance barely changes the fit; only the animal-size covariate moves the score.

Design	pseudo-R^2	LSO Δ (p)	LLO Δ (p)
full (4 distances)	0.055	+0.004 (p=0.24)	+0.004 (p=0.23)
drop taxonomic	0.039	+0.000 (p=0.47)	+0.000 (p=0.49)
drop temporal	0.055	+0.004 (p=0.24)	+0.004 (p=0.23)
drop visual	0.048	+0.004 (p=0.22)	+0.004 (p=0.28)
drop environment	0.055	+0.006 (p=0.15)	+0.005 (p=0.17)
+ animal size	0.119	+0.018 (p=0.02)	+0.020 (p=0.00)
− support covariate	0.035	−0.002 (p=0.72)	−0.003 (p=0.73)

Table B.5: Design-robustness sweep (detection). Each row re-runs the primary model with one design choice swapped; entries are the out-of-sample MAE gain, Δ [95% CI] (p), under leave-species-out (LSO) and leave-location-out (LLO) (Bonferroni $m = 5$). The null holds throughout — no encoder, crop, prompt, or distributional distance beats the mean.

Variant	LSO Δ [CI] (p)	LLO Δ [CI] (p)	Beats mean?
baseline (DINOv2, mask-crop, species prompt)	+0.004 [-0.006, +0.015] (0.237)	+0.004 [-0.007, +0.015] (0.235)	n.s.
CLIP encoder	+0.004 [-0.006, +0.014] (0.247)	+0.004 [-0.007, +0.014] (0.270)	n.s.
whole-frame (bbox + background)	+0.002 [-0.008, +0.013] (0.349)	+0.002 [-0.008, +0.013] (0.357)	n.s.
generic "animal" prompt	-0.346 [-0.381, -0.309] (1.000)	-0.343 [-0.363, -0.326] (1.000)	n.s.
Frechet distributional distance	+0.007 [-0.005, +0.019] (0.120)	+0.007 [-0.004, +0.019] (0.102)	n.s.
MMD distributional distance	+0.002 [-0.008, +0.012] (0.374)	+0.002 [-0.009, +0.012] (0.349)	n.s.

Table B.6: SAM 3 familiarity proxy (detection): does a species' separability in SAM 3's own feature space predict `pDetA`? Each metric is size-controlled; ΔMAE (p) is the out-of-sample gain over a mean predictor. None clears Bonferroni ($\alpha = 0.017$) on either dataset, and all three re-encode the visual distance ($r_{\text{vis}} = 0.63$–0.85). As the size control validates on the same cells, this is a genuine null, not an underpowered one.

Metric	SA-FARI ΔMAE (p)	BURST ΔMAE (p)	r_{vis}	r_{size}	clears bar?
silhouette	+0.012 (0.034)	+0.012 (0.020)	+0.67	−0.16	no
nearest-prototype	+0.009 (0.068)	+0.006 (0.136)	+0.63	+0.03	no
mahalanobis	+0.008 (0.085)	+0.006 (0.148)	+0.85	+0.32	no

Table B.7: Cross-model swap on BURST: out-of-sample error of the combined four-distance model (leave-species-out, size-controlled) vs a mean predictor, per challenger; ΔMAE with a 95% CI and one-sided p (Bonferroni $\alpha = 0.0125$). Both challengers track BURST animals (Table 4.9) yet the distance model is null for each — the label-free distance null replicates across trackers.

Challenger	ΔMAE	95% CI	p
GLEE	+0.011	[−0.004, +0.027]	0.088
Florence-2 + SAM 2	+0.034	[−0.001, +0.068]	0.030

Table B.8: BURST independent replication — out-of-sample error under leave-species-out cross-validation (41 species, 132 mask-native cells) vs a mean predictor. ΔMAE is the `pDetA` gain over baseline, with a paired-bootstrap 95% CI and one-sided p. Every distance is null; not even the animal-size control reaches significance, so BURST is a convergent null that cannot certify power for a small species-level effect.

Predictor	ΔMAE	95% CI	p
distances (taxonomic + visual + environment)	+0.002	[−0.007, +0.011]	0.374
visual only	+0.003	[−0.005, +0.013]	0.257
taxonomic only (full-taxonomy subset, n=82)	−0.006	—	0.952
environment only	−0.001	[−0.005, +0.003]	0.747
animal size (log-area) [control]	+0.005	[−0.003, +0.014]	0.143